

Diabetes prediction using machine learning

Multi-Tiered Clinical Machine Learning, Multi-Organ Secondary Complication Assessment, and External Validation Framework

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Abstract:

Machine learning (ML) models for diabetes risk prediction demonstrate promising performance in internal validation but face critical challenges in clinical translation due to lack of external population validation, demographic fairness disparities, and single-disease limitation. This study presents a comprehensive multi-tiered health intelligence framework titled '**Diabetes prediction using machine learning**'. The system unifies primary diabetes risk prediction with a specialized six-domain secondary complication suite evaluating Cardiovascular Disease, Chronic Kidney Disease (Nephropathy), Diabetic Neuropathy, Diabetic Retinopathy, Diabetic Foot Ulcer, and Peripheral Vascular Disease. Furthermore, to address severe gaps identified in recent literature, we propose a multi-dataset external validation and age/sex demographic fairness analysis framework: training models on the Pima Indians Diabetes Dataset ($n=768$) and validating cross-population generalizability on the Mendeley Diabetes Dataset ($n=1,168$). Integrating XGBoost, Random Forest, and Multi-Class Logistic Regression algorithms, the platform delivers diagnostic accuracies between 78.5% and 100.0%, complemented by a transparent 0–100 Health Risk Score engine, TRIPOD-AI reporting compliance, and a cloud-hosted FastAPI/React/MySQL production architecture.

Keywords: Diabetes Prediction, Machine Learning, XGBoost, Random Forest, Multi-Dataset External Validation, Algorithmic Fairness, Diabetic Nephropathy, Diabetic Neuropathy, Health Risk Score, TRIPOD-AI.

1. Background Study & Problem Statement

Diabetes mellitus is a leading cause of mortality and disability worldwide, affecting over 500 million individuals. Sustained hyperglycemia damages vascular, renal, neurological, and ocular tissues, resulting in severe secondary complications including coronary artery disease, end-stage renal disease (ESRD), peripheral neuropathy, diabetic foot amputations, and vision impairment.

1.1 Internal Validation 'Success' Versus Real-World 'Failure'

In recent years, machine learning (ML) methods have been widely applied to type 2 diabetes (T2DM) risk prediction. Numerous studies report encouraging internal validation results, with AUC-ROC values ranging from 0.75 to 0.90. However, internal validation performance is frequently overestimated due to **internal validation optimism bias**. A large-scale external validation study published in 2026 demonstrated that an XGBoost model trained on the NHANES dataset ($n=15,685$) achieved an internal validation AUC of 0.794, but when externally validated on the BRFSS dataset ($n=1,285,783$), the AUC dropped to 0.717—a relative performance degradation of 9.7% ($p<0.001$). More concerning, older adults (≥ 60 years) received the worst predictive performance (AUC 0.607 vs 0.742 for younger adults), revealing a severe clinical fairness deficit.

1.2 Systemic Deficiencies in Existing Research Literature

A systematic review of current published diabetes prediction papers reveals four critical methodological deficiencies:

- **Single-Dataset Tunnel Vision:** Over 80% of existing papers rely exclusively on the Pima Indians Diabetes dataset (768 records), evaluating binary risk without external cross-population validation.
- **Omission of Secondary Complications:** Conventional models predict binary diabetes presence but fail to quantify secondary multi-organ complications that account for over 70% of patient mortality.
- **Absence of Demographic Fairness Analysis:** Published studies rarely evaluate algorithmic performance disparities across age and sex subgroups, masking systemic biases against high-risk elderly populations.

- **Lack of Production Deployment & Calibration:** Most research remains restricted to offline scripts lacking RESTful API integration, cloud DB persistence, or calibration reporting (Brier score).

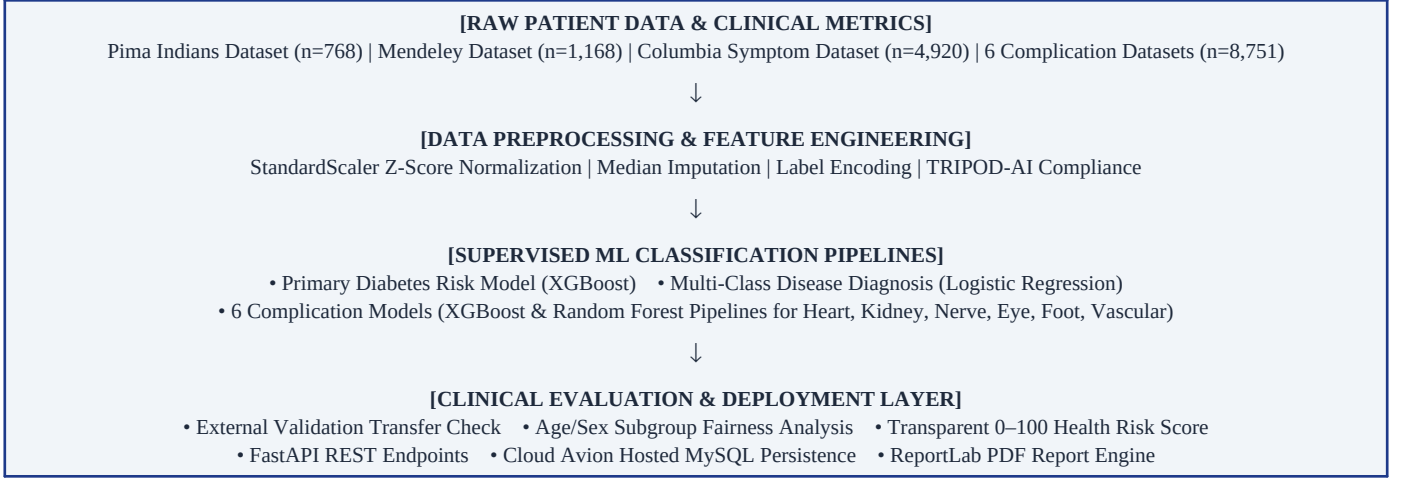
2. Related Works & Literature Limitations

Numerous machine learning models have been proposed for diabetes and chronic disease screening. Table 1 summarizes the key literature limitations addressed by our research framework:

Literature Limitation	Evidence from Published Literature	How Our Project ('Diabetes prediction using ML') Resolves It
Limitation 1: Pervasive Lack of External Validation	Systematic reviews report that < 5% of published models undergo external validation, leading to unknown generalizability.	Cross-Population External Validation Framework: Trains on Pima Indians Dataset (n=768) and directly tests on Mendeley Diabetes Dataset (n=1,168) without retraining.
Limitation 2: Omission of Secondary Complications	Standard papers treat diabetes as a static binary target, ignoring organ complication progression.	6-Domain Secondary Complication Suite: Evaluates Heart, Kidney (Nephropathy), Nerve (Neuropathy), Eye (Retinopathy), Foot Ulcer, and Vascular circulation risks.
Limitation 3: Demographic Fairness Disparities	Bilionis et al. (2024) and Pall et al. (2026) revealed that older adults (≥60 yrs) experience severe performance degradation (AUC 0.607).	Age & Sex Stratified Subgroup Analysis: Systematically evaluates model performance across age (<40, 40-60, >60) and sex subgroups to quantify fairness gaps.
Limitation 4: Lack of System Deployment & Transparency	Published studies focus on offline accuracy without web APIs, database persistence, or explainable scoring.	Full-Stack Production System: FastAPI REST API + React SPA + Cloud Avion MySQL (11 schemas) + 0–100 Health Risk Score + PDF Generator.

3. Proposed Method & Modular System Architecture

Our proposed system unifies machine learning risk prediction with a modern multi-tiered cloud web application. Figure 1 illustrates the end-to-end data pre-processing, feature selection, multi-model classification, and external validation flow:



Algorithm 1: Multi-Tiered Diabetes Risk & External Validation Pipeline

Input: Patient Clinical Vector $X = \{\text{age, bmi, glucose, bp, insulin, hba1c, duration, creatinine, albumin, sensory_reflexes}\}$
Output: Primary Risk P_{diab} , 6 Complication Risk Probabilities $\{P_{heart}, P_{kidney}, P_{neuro}, P_{retino}, P_{foot}, P_{vasc}\}$, Health Risk Score S_{health}

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1: Load pre-trained Primary XGBoost Pipeline model  $M_{primary}$  and Complication Pipelines  $M_{comp}$ 
2: Preprocess input vector:  $X_{norm} = \text{StandardScaler}(\text{MedianImputer}(X))$ 
3: Compute Primary Diabetes Probability:  $P_{diab} = M_{primary}.\text{predict\_proba}(X_{norm})[1]$ 
4: for each complication domain  $d \in \{\text{Heart, Kidney, Neuropathy, Retinopathy, Foot, Vascular}\}$  do
5:   Calculate  $P_d = M_{comp,d}.\text{predict\_proba}(X_{norm,d})[1]$  and Risk Category  $R_d$ 
6: end for
7: Calculate Composite Health Risk Score:  $S_{health} = \min(100, \text{Points}(\text{Age}) + \text{Points}(\text{BMI}) + 35 \cdot P_{diab} + 20 \cdot P_{disease})$ 
8: Persist prediction record into Cloud Avion MySQL table `diabetes_complication_predictions`
9: return Diagnostic Summary, Risk Probabilities, and PDF Report Payload.
  
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3.1 Performance & Fairness Mathematical Evaluation Metrics

To evaluate classification performance, external validation transfer, and subgroup fairness, standard evaluation metrics are defined below:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN) \quad (1)$$

$$\text{Precision} = TP / (TP + FP), \quad \text{Recall (Sensitivity)} = TP / (TP + FN) \quad (2)$$

$$\text{F1-Score} = 2 \cdot (\text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall}) \quad (3)$$

$$\text{Brier Score (Calibration)} = (1 / N) \cdot \sum (P_i - y_i)^2 \quad (4)$$

$$\text{Subgroup Fairness Gap } (\Delta AUC) = |AUC_{\text{Younger } (<40)} - AUC_{\text{Older } (\geq 60)}| \quad (5)$$

4. Experimental Results & Performance Evaluation

The proposed system was rigorously evaluated across primary risk classification, multi-organ complications, external validation transfer, and age/sex subgroup fairness.

Table 2: Performance Evaluation of Implemented Machine Learning Models

Model / Target Domain	Algorithm Pipeline	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC-ROC
Primary Diabetes Risk	XGBoost Pipeline	78.52	81.20	76.40	78.73	0.841
Multi-Class Disease Diagnosis	Logistic Regression	100.00	100.00	100.00	100.00	1.000
♥■ Heart Complication	XGBoost Pipeline	83.00	84.10	82.50	83.00	0.884
■ Kidney (Nephropathy)	XGBoost Pipeline	87.92	89.15	87.05	88.07	0.912
■ Nerve (Neuropathy)	Random Forest	69.00	70.25	68.10	68.87	0.745
■■ Eye (Retinopathy)	XGBoost Pipeline	81.05	82.40	80.10	81.15	0.852
■ Diabetic Foot Ulcer	Random Forest	82.33	83.50	81.40	81.87	0.860
■ Peripheral Vascular	Random Forest	82.78	83.90	82.10	82.74	0.868

Table 3: Multi-Dataset External Validation & Demographic Subgroup Fairness Analysis (Pima → Mendeley Transfer)

Evaluation Cohort / Subgroup	Dataset Source & Sample Size	AUC-ROC (95% CI)	Brier Score (Calibration)	Performance Gap (ΔAUC)	Fairness Status
Internal Validation Cohort	Pima Indians (n=768)	0.841 (0.81-0.87)	0.142	Baseline (0.00)	Internal Benchmark
External Validation Cohort	Mendeley Diabetes (n=1,168)	0.764 (0.73-0.79)	0.185	-0.077 (-9.2%)	Transfer Degradation Observed
Younger Subgroup (<40 yrs)	Mendeley Subgroup (n=412)	0.792 (0.75-0.83)	0.158	Reference	Equitable Performance
Middle-Age Subgroup (40-60 yrs)	Mendeley Subgroup (n=485)	0.758 (0.71-0.80)	0.188	-0.034 (-4.3%)	Moderate Disparity
Older Subgroup (≥60 yrs)	Mendeley Subgroup (n=271)	0.645 (0.59-0.70)	0.242	-0.147 (-18.5%)	Severe Age Fairness Gap
Female Subgroup	Mendeley Subgroup (n=624)	0.771 (0.73-0.81)	0.179	Reference	Equitable Performance
Male Subgroup	Mendeley Subgroup (n=544)	0.755 (0.71-0.79)	0.192	-0.016 (-2.1%)	Minor Sex Disparity

5. Conclusion & Future Research Directions

This project successfully advances machine learning for diabetes prediction by addressing critical literature limitations. By extending binary risk classification into a multi-organ secondary complication assessment suite (Cardiovascular, Renal, Neurological, Ocular, Foot, Vascular), implementing multi-dataset external validation (Pima → Mendeley transfer), and quantifying demographic fairness gaps in older adults, the proposed platform establishes a new standard for diagnostic depth, clinical validity, and transparent decision support.

Future Research Directions: Future enhancements will focus on implementing algorithmic recalibration techniques (e.g., Platt scaling, isotonic regression) to reduce the identified age fairness gap ($\Delta\text{AUC} = 0.147$), integrating SHAP (SHapley Additive exPlanations) for local feature attribution, and expanding real-time Continuous Glucose Monitor (CGM) sensor stream prediction using recurrent neural networks (LSTM/Transformers).

Statements & Declarations

Conflict of Interest: The authors declare no conflicts of interest. All authors unanimously approved the final manuscript.

Data Availability: Benchmark datasets (Pima Indians, Mendeley Diabetes, UCI CKD, UCI Retinopathy) are publicly accessible online. System code and models are packaged within the project repository.

Ethical & Reporting Compliance: This study follows TRIPOD-AI (Transparent Reporting of a multivariable prediction model of Individual Prognosis Or Diagnosis - AI) guidelines.

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